

**BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)**

**CLASS: BTECH/IMSC
BRANCH: BT/CE/CHEMICAL/ME/PIE/FT/PHYSICS**

**SEMESTER: I/ADD
SESSION : MO/2024**

SUBJECT: BE24101 BIOLOGICAL SCIENCE FOR ENGINEERS

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
 2. Attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates
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		CO	BL
Q.1(a)	Write the characteristic features of living organisms.	[2]	1 2
Q.1(b)	Give a brief illustration on any two theories of origin of life	[3]	1 1
Q.2(a)	Why is biology important for engineers?	[2]	1 3
Q.2(b)	How will you differentiate between Plant and Animal cells? Explain with their contrasting features.	[3]	2 4
Q.3(a)	What is a cell? How does the cell theory establish the importance of cell?	[2]	2 4
Q.3(b)	How can genome editing be useful in solving real world problem?	[3]	3 3
Q.4(a)	Describe the basic structure of protein.	[2]	2 1
Q.4(b)	What is central dogma of molecular biology? Why it is important to understand?	[3]	3 3
Q.5(a)	What is DNA? Briefly describe its structure	[2]	2 2
Q.5(b)	Explain the law of segregation in Mendelian genetics.	[3]	2 3

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